

Recent Advancements in a Posteriori Error Estimation for Algebraic Stabilizations

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1 Algebraic Stabilization Schemes

2 A Posteriori Error Analysis

3 Numerical Studies

4 Conclusion & Outlook



- Steady-state convection-diffusion-reaction equation

$$\begin{aligned} -\varepsilon \Delta u + \mathbf{b} \cdot \nabla u + cu &= f \quad \text{in } \Omega, \\ u &= u_D \quad \text{on } \Gamma_D, \\ -\varepsilon \nabla u \cdot \mathbf{n} &= u_N \quad \text{on } \Gamma_N \end{aligned}$$

- Ω – bounded polyhedral Lipschitz domain in $\mathbb{R}^d$, $d \in \{2, 3\}$
- $\mathbf{n}$ – outward pointing unit normal
- Assume

$$\left(c(x) - \frac{1}{2} \nabla \cdot \mathbf{b}(x) \right) \geq \sigma \geq 0$$

- Interested in convection-dominated regime, $\varepsilon \ll \|\mathbf{b}\|_{L^\infty(\Omega)} L$
 - L – Characteristic length of the problem



- Ideal discretization
 1. Accurate and sharp layers¹

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- Idea: Combine both the approaches

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- Fully Computable Error Bound
 - Implicit Residual Based Estimator¹
 - Local Efficiency depends on Linearity Preserving

¹ Allendes, Barrenechea, Rankin: *SIAM Journal on Scientific Computing*, 39 (5), A1903–A1927, 2017

² J.: *Computers & Mathematics with Applications*, 97 (1), 86–99, 2021

³ J., John, Knobloch: *SIAM Journal on Scientific Computing*, 45 (4), B564–B589, 2023



- Fully Computable Error Bound
 - Implicit Residual Based Estimator¹
 - Local Efficiency **depends** on Linearity Preserving
- Residual Based Error Estimator²
 - Explicit Residual Based Estimator
 - **Formal** Local Lower Bound
 - Hanging Nodes Theory³

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- Variational problem for AS scheme

Find $u_h \in V_h$ such that

$$a_h(u_h, v_h) + b_h(u_h; u_h, v_h) = \langle f, v_h \rangle \quad \forall v_h \in V_h$$

- V_h – finite element space with homogeneous Dirichlet boundary conditions ($V_h \subset V$)

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$$b_h(w; z, v) = \sum_{i,j=1}^N b_{ij}(w)(z_j - z_i)v_i \quad \forall w, z, v \in V_h$$

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- Another representation of stabilization for $w, v, z \in V_h$,¹

$$b_h(w; z, v) = \sum_{E \in \mathcal{E}_h} |b_E(w)| h_E (\nabla z \cdot \mathbf{t}_E, \nabla v \cdot \mathbf{t}_E)$$

¹Barrenechea, John, Knobloch, Rankin: *SeMA Journal*, 75 (4), 655–685, 2018



- AS norm

$$\|u_h\|_{AS}^2 = \|u_h\|_a^2 + b_h(u_h, u_h, u_h) \quad \forall u_h \in V_h$$

- where $\|u_h\|_a^2 = \varepsilon |u_h|_1^2 + \sigma \|u_h\|_0^2$

¹ John, Novo: *Computer Methods in Applied Mechanics and Engineering*, 255 (1), 289–305, 2013



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- Let $\mathcal{I}_h u$ denote the Scott-Zhang interpolation operator. Galerkin orthogonality arguments

$$\begin{aligned} \|u - u_h\|_{AS}^2 &= \langle f, u - \mathcal{I}_h u \rangle + \langle u_N, u - \mathcal{I}_h u \rangle_{\Gamma_N} - a_h(u_h, u - \mathcal{I}_h u) \\ &\quad + b_h(u_h; u, \mathcal{I}_h u - u_h) \end{aligned}$$

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- Standard residual a posteriori error bound ¹

$$\begin{aligned} &\langle f, u - \mathcal{I}_h u \rangle + \langle u_N, u - \mathcal{I}_h u \rangle_{\Gamma_N} - a_h(u_h, u - \mathcal{I}_h u) \\ &= \sum_{K \in \mathcal{T}_h} (R_K(u_h), u - \mathcal{I}_h u)_K + \sum_{F \in \mathcal{F}_h} \langle R_F(u_h), u - \mathcal{I}_h u \rangle_F \end{aligned}$$

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with

$$\begin{aligned} R_K(u_h) &:= f + \varepsilon \Delta u_h - \mathbf{b} \cdot \nabla u_h - c u_h|_K, \\ R_F(u_h) &:= \begin{cases} -\varepsilon \llbracket \nabla u_h \cdot \mathbf{n}_F \rrbracket_F & \text{if } F \in \mathcal{F}_{h,\Omega}, \\ u_N - \varepsilon (\nabla u_h \cdot \mathbf{n}_F) & \text{if } F \in \mathcal{F}_{h,N}, \\ 0 & \text{if } F \in \mathcal{F}_{h,D} \end{cases} \end{aligned}$$



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- Using interpolation estimates, Cauchy-Schwarz, and Young's inequality

$$\begin{aligned} & \|u - u_h\|_a^2 + \frac{C_Y}{C_Y - 1} b_h(u_h; u - u_h, u - u_h) \\ & \leq \frac{C_Y^2}{2(C_Y - 1)} \sum_{K \in \mathcal{T}_h} \min \left\{ \frac{C_I^2}{\sigma}, \frac{C_I^2 h_K^2}{\varepsilon} \right\} \|R_K(u_h)\|_{L^2(K)}^2 \\ & \quad + \frac{C_Y^2}{2(C_Y - 1)} \sum_{F \in \mathcal{F}_h} \min \left\{ \frac{C_F^2 h_F}{\varepsilon}, \frac{C_F^2}{\sigma^{1/2} \varepsilon^{1/2}} \right\} \|R_F(u_h)\|_{L^2(F)}^2 \\ & \quad + \frac{C_Y}{C_Y - 1} b_h(u_h; u, \mathcal{I}_h u - u_h) \end{aligned}$$



- Linearity of $b_h(\cdot; \cdot, \cdot)$,

$$b_h(u_h; u, \mathcal{I}_h u - u_h) = b_h(u_h; u - u_h, \mathcal{I}_h u - u_h) + b_h(u_h; u_h, \mathcal{I}_h u - u_h)$$



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- Using interpolation estimates, Cauchy-Schwarz, trace inequality, inverse estimate, and Young's inequality

$$\begin{aligned} b_h(u_h; u_h, \mathcal{I}_h u - u_h) &\leq \frac{C_Y}{2} \sum_{E \in \mathcal{E}_h} \min \left\{ \frac{\kappa_1 h_E^2}{\varepsilon}, \frac{\kappa_2}{\sigma} \right\} |b_E|^2 h_E^{1-d} \\ &\quad \times \|\nabla u_h \cdot \mathbf{t}_E\|_{L^2(E)}^2 + \frac{1}{C_Y} \|u - u_h\|_a^2, \end{aligned}$$

where

$$\begin{aligned} \kappa_1 &= C_{\text{edge}, \max} \left(1 + (1 + C_{\mathcal{I}})^2 \right), \\ \kappa_2 &= C_{\text{inv}}^2 C_{\text{edge}, \max} \left(1 + (1 + C_{\mathcal{I}})^2 \right). \end{aligned}$$



Theorem (Global a Posteriori Error Estimate)

A global a posteriori error estimate for the energy norm is given by^{1,2}

$$\|u - u_h\|_a^2 \leq \eta_1^2 + \eta_2^2 + \eta_3^2,$$

where

$$\eta_1^2 = \sum_{K \in \mathcal{T}_h} \min \left\{ \frac{4C_{\mathcal{I}}^2}{\sigma}, \frac{4C_{\mathcal{I}}^2 h_K^2}{\varepsilon} \right\} \|R_K(u_h)\|_{L^2(K)}^2,$$

$$\eta_2^2 = \sum_{F \in \mathcal{F}_h} \min \left\{ \frac{4C_F^2 h_F}{\varepsilon}, \frac{4C_F^2}{\sigma^{1/2} \varepsilon^{1/2}} \right\} \|R_F(u_h)\|_{L^2(F)}^2,$$

$$\eta_3^2 = \sum_{E \in \mathcal{E}_h} \min \left\{ \frac{4\kappa_1 h_E^2}{\varepsilon}, \frac{4\kappa_2}{\sigma} \right\} |b_E|^2 h_E^{1-d} \|\nabla u_h \cdot \mathbf{t}_E\|_{L^2(E)}^2$$

¹ J.: *Computers & Mathematics with Applications*, 97 (1), 86–99, 2021

² J.: *Applied Mathematics Letters*, 157 (1), 109192, 2024



- Algebraic Flux Correction Schemes
 - Stabilization Term

$$b_h(u; v, w) = \sum_{i,j=1}^N (1 - \alpha_{ij}(u)) d_{ij} (v(x_j) - v(x_i)) w(x_i)$$

¹ Barrenechea, John, Knobloch: *Mathematical Models and Methods in Applied Sciences*, 27 (3), 525–548, 2017

² Kuzmin: *Computer Methods in Applied Mechanics and Engineering*, 361 (1), 112804, 2020



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- $d_{ij} = \begin{cases} -\max\{a_{ij}, 0, a_{ji}\}, & i \neq j \\ -\sum_{i \neq j} d_{ij} & i = j \end{cases}$

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- $d_{ij} = \begin{cases} -\max\{a_{ij}, 0, a_{ji}\}, & i \neq j \\ -\sum_{i \neq j} d_{ij} & i = j \end{cases}$
 - Limiters (Symmetric)
 - BJK¹
 - Monolithic Convex²

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² Barrenechea, John, Knobloch: *Springer*, 61 (1), 441–447, 2025

³ John, Knobloch: *Numerische Mathematik*, 152 (3), 553–585, 2022

⁴ Knobloch: *Numerical Algorithms*, 94 (2), 547–580, 2023



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- Limiters

- **BBK**^{1,2} ($b_{ij} = -d_{ij} \max \{ \eta_i(u), \eta_j(u) \}^p$)
- **MUAS**³
- **SMUAS**⁴

¹ Barrenechea, Burman, Karakatsani: *Numerische Mathematik*, 135 (2), 521–545, 2017

² Barrenechea, John, Knobloch: *Springer*, 61 (1), 441–447, 2025

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 - Effectivity Index in $\| \cdot \|_a (\gamma_{\text{eff}})$



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 - Conforming Closure
 - Nonlinear Equations



- Iterative solver

- Matrix formulation of the algebraic stabilised schemes^{1,2}

$$(\mathbb{A} + \mathbb{D}) \mathbf{U} = \mathbf{F} + (\mathbb{D} - \mathbb{B}(\mathbf{U})) \mathbf{U}$$

- Fixed point right-hand side

$$\begin{aligned} (\mathbb{A} + \mathbb{D}) \tilde{\mathbf{U}}^\mu &= \mathbf{F} + (\mathbb{D} - \mathbb{B}(\mathbf{U}^\mu)) \mathbf{U}^\mu, \\ \mathbf{U}^{\mu+1} &= \omega \tilde{\mathbf{U}}^\mu + (1 - \omega) \mathbf{U}^\mu, \end{aligned}$$

where $1 \geq \omega > 0$ is a dynamic damping parameter³

¹ J., John: *BAIL* 2018, 135 (1), 113–128, 2020

² J., John: *Computers & Mathematics with Applications*, 78 (9), 3117–3138, 2019

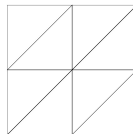
³ John, Knobloch: *Computer Methods in Applied Mechanics and Engineering*, 196 (17-20), 2197–2215, 2007



- **Corner Boundary Layer**¹

- $\Omega = (0, 1)^2$, $\varepsilon = 10^{-2}$, $\mathbf{b} = (2, 3)^\top$, $\mathbf{c} = 1$, $u_D = 0$, $u_N = 0$, $\Gamma = \Gamma_D$, and f such that
-

$$u(x, y) = xy^2 - y^2 e^{\left(\frac{2(x-1)}{\varepsilon}\right)} - x e^{\left(\frac{3(y-1)}{\varepsilon}\right)} + e^{\left(\frac{2(x-1)+3(y-1)}{\varepsilon}\right)}$$



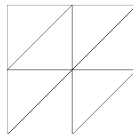
¹ John, Knobloch, Savescu: *Computer Methods in Applied Mechanics and Engineering*, 200 (**41-44**), 2916–2929, 2011

² J., John: *Computers & Mathematics with Applications*, 78 (**1**), 3117–3138, 2019

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- stop of the non linear iteration²

- 10000 iterations
 - $\|\text{residual}\|_{\ell^2} \leq \sqrt{\text{\#dof}} 10^{-8}$

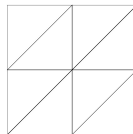
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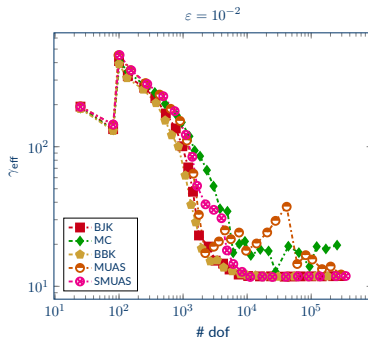


- stop of the non linear iteration²
 - 10000 iterations
 - $\|\text{residual}\|_{\ell^2} \leq \sqrt{\text{\#dof}} 10^{-8}$
- stop of the adaptive algorithm
 - $\eta \leq 10^{-3}$
 - $\text{\#dof} \approx 2.5 \times 10^6$

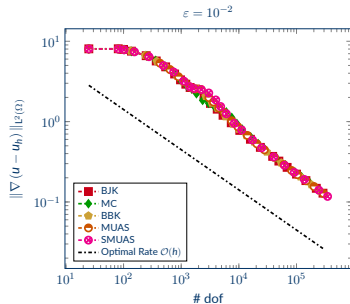
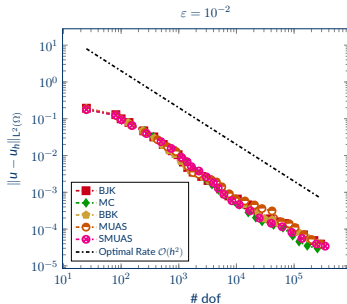
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- Effectivity Index

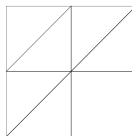
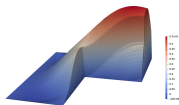


- $L^2(\Omega)$ Error and $L^2(\Omega)$ Error of the gradient



- **L-Shaped Domain**¹

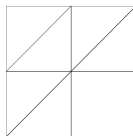
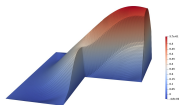
- $\Omega = (0, 1)^2 \setminus (0.5, 1) \times (0, 0.5)$, $\epsilon = 10^{-6}$, $\mathbf{b} = (3, 1)^\top$, $\mathbf{c} = 1$, and $\Gamma = \Gamma_D$
- $f = 100r(r - 0.5)(r - 1/\sqrt{2})$, $u_D = 0$
- $u \notin H^2(\Omega)$



¹ John: *Computer Methods in Applied Mechanics and Engineering*, 190 (5-7), 757–781, 2000

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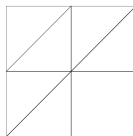
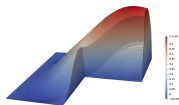


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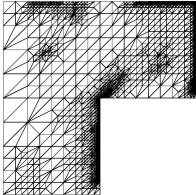
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- $f = 100r(r - 0.5)(r - 1/\sqrt{2})$, $u_D = 0$
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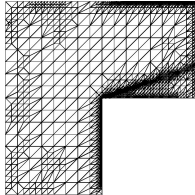
- stop of the non linear iteration
 - 10000 iterations
 - $\|\text{residual}\|_{\ell^2} \leq \sqrt{\text{\#dof}} 10^{-8}$
- stop of the adaptive algorithm
 - $\eta \leq 10^{-3}$
 - $\text{\#dof} \approx 5 \times 10^5$

¹ John: *Computer Methods in Applied Mechanics and Engineering*, 190 (5-7), 757–781, 2000

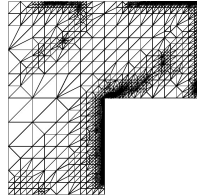
- Adaptive Grids ($\#dof = 2 \times 10^5$)



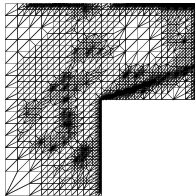
(a) BJK



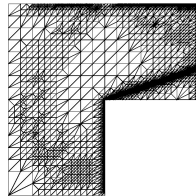
(b) Monolithic



(c) BBK

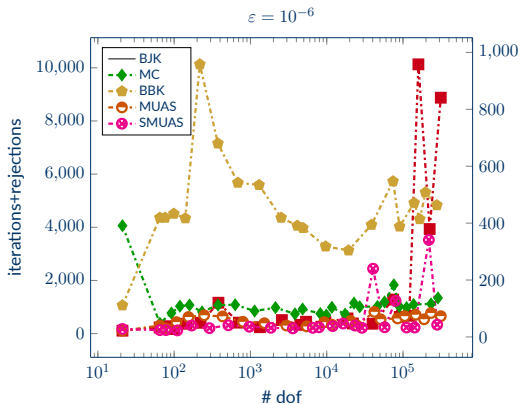


(d) MUAS



(e) SMUAS

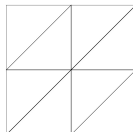
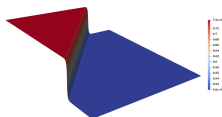
- Efficiency



- Nonlinear Convection¹

- $\Omega = (0, 1)$, $\varepsilon = 10^{-3}$, $\mathbf{b} = (u, u)^\top$, $\mathbf{c} = 0$, and $\Gamma = \Gamma_D$
 - f and u_D are chosen such that
 -

$$u(x, y) = \frac{3}{4} - \frac{1}{4 \left[1 + \exp \left(\frac{-4x+4y-1}{32\varepsilon} \right) \right]}$$

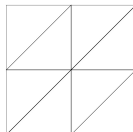
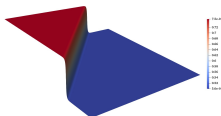


¹Zhao, Li, Gao: *Journal of Mathematical Analysis and Applications*, 550 (2), 129617, 2025

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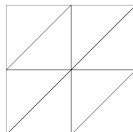
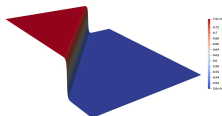
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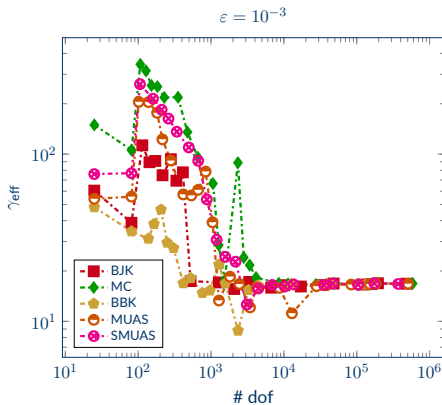
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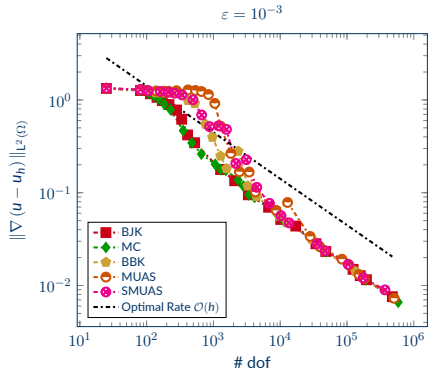
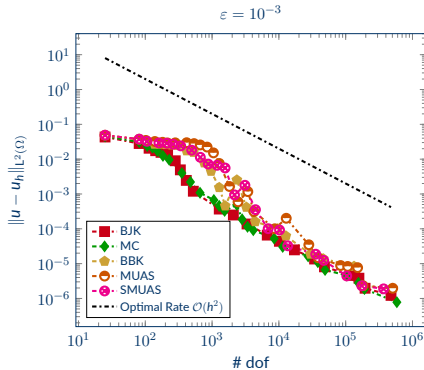
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¹ Zhao, Li, Gao: *Journal of Mathematical Analysis and Applications*, 550 (2), 129617, 2025

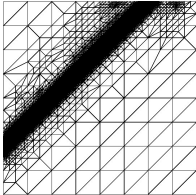
- Effectivity Index



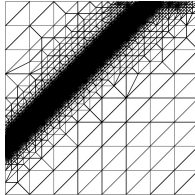
- $L^2(\Omega)$ Error and $L^2(\Omega)$ Error of the gradient



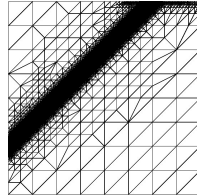
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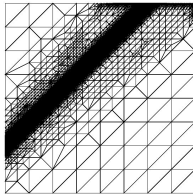
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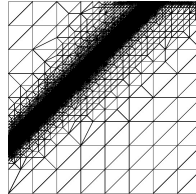
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 - MC achieves the most balanced combination of accuracy, γ_{eff} , and mesh adaptation.
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 - **MC** achieves the most balanced combination of **accuracy**, γ_{eff} , and **mesh adaptation**.
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- **Nonlinear Problems**
 - The behavior changes significantly for solution-dependent transport
 - **BJK** produces the most reliable adaptive meshes and estimator behavior.
 - No single limiter is uniformly superior for all quantities of interest
- **Overall Observation**
 - The interaction between **stabilization**, **error estimation**, and **adaptive refinement** is strongly problem dependent.



- Summary¹
 - Linear problems: AFC+MC gives the most balanced performance
 - Nonlinear problems: AFC+BJK is the most reliable choice for error estimation and adaptive refinement
 - MUAS provides an efficient alternative with competitive accuracy
 - Numerical results indicate that the effectiveness of AFC/AS methods depends strongly on the problem type and adaptive mesh evolution

¹ Ahmed, J., arXiv: 2602.14504, 2026

